

# Discrete Logarithm Algorithm in Diffie-Hellman Attacks: A Quantum Computing Approach

## 1 Introduction

The Discrete Logarithm Algorithm (DLA) plays a crucial role in cryptanalysis, particularly in attacking the Diffie-Hellman key exchange protocol. This paper examines how DLA can be exploited to compromise the security of Diffie-Hellman and presents a quantum algorithm that provides a polynomial-time solution using the Quantum Fourier Transform (QFT).

## 2 DLA in Diffie-Hellman Attack

The Diffie-Hellman key exchange protocol relies on the computational hardness of the discrete logarithm problem. Given a prime modulus  $p$ , a generator  $g$ , and public values  $g^a \bmod p$  and  $g^b \bmod p$ , an attacker who can efficiently solve the discrete logarithm problem can recover:

$$a = \log_g(g^a \bmod p) \tag{1}$$

With knowledge of  $a$ , the attacker can then compute the shared secret:

$$k = (g^b)^a \bmod p \tag{2}$$

The classical DLA has sub-exponential complexity, making it computationally infeasible for large prime moduli. However, quantum computing offers a polynomial-time solution.

### 3 Quantum Algorithm for DLA

We present a quantum algorithm based on the QFT that solves the discrete logarithm problem in polynomial time. The algorithm follows these steps:

#### 3.1 State Preparation

Initialize two quantum registers in the state:

$$|\psi_0\rangle = \frac{1}{\sqrt{p}} \sum_{x=0}^{p-1} |x\rangle |0\rangle \quad (3)$$

#### 3.2 Function Evaluation

Apply the modular exponentiation function:

$$|\psi_1\rangle = \frac{1}{\sqrt{p}} \sum_{x=0}^{p-1} |x\rangle |g^x \bmod p\rangle \quad (4)$$

#### 3.3 Quantum Fourier Transform

Apply the QFT to the first register:

$$QFT : |x\rangle \rightarrow \frac{1}{\sqrt{p}} \sum_{y=0}^{p-1} e^{2\pi i xy/p} |y\rangle \quad (5)$$

The resulting state becomes:

$$|\psi_2\rangle = \frac{1}{p} \sum_{x=0}^{p-1} \sum_{y=0}^{p-1} e^{2\pi i xy/p} |y\rangle |g^x \bmod p\rangle \quad (6)$$

#### 3.4 Measurement

Measure both registers. The period finding aspect of the QFT allows us to extract the discrete logarithm with high probability.

## 4 Complexity Analysis

The quantum algorithm achieves polynomial time complexity through:

1. State preparation:  $O(\log p)$  qubits and operations
2. Modular exponentiation:  $O(\log^3 p)$  operations
3. QFT implementation:  $O(\log^2 p)$  operations

The total complexity is  $O(\log^3 p)$ , which is exponentially faster than the best classical algorithms.

## 5 Conclusion

The quantum solution to the discrete logarithm problem demonstrates the potential vulnerability of classical cryptographic systems to quantum attacks. This has significant implications for the security of Diffie-Hellman and similar protocols in a post-quantum world, necessitating the development of quantum-resistant alternatives.